



UNIVERSITY OF AMSTERDAM

Digital Marketing & Analytics:

Introduction, Platforms, Data Types, Metrics & Conversion-Attribution I

Dr. Umut Konuş

Digital Marketing and Analytics: Week 1

Academic Year 2025-2026 – Semester 1 – Block 2

**AMSTERDAM
BUSINESS
SCHOOL**

Business

Practical Introduction

Lectures and Workgroups

Lectures: **Fridays, 09:00-11:00 – IN-CLASS (ON CAMPUS)**

The lectures & workgroups are LIVE & on-campus.

Slides are shared 2-3 days prior to class on CANVAS.

For Tutorials 1-2 some practical content (recorded) will be shared before (or more likely after) these tutorials.

You are expected to watch these videos (knowledge clips), so you can easily catch-up with the practical applications now and later.



* Please check on Rooster/CANVAS which workgroup you are allocated by the system. You have to participate in the workgroup you are allocated in.

Assessment & Grading

(1) Final Exam: 60% minimum 5,5. 2 hours on a designated time-slot.

Open (and/ some Multiple-Choice Questions) – based on slides/learnings.

NO analytical-software practice during exam ! Analytical questions will be doable by pen-paper or interpretation of analysis findings/model results.

(2) Main Team Assignments: 40% (each 20%) – minimum 5,5 each

(3) Mini Team Assignments: 1 data (mini) exercise/assignment per group

- *Mini-assignments have to be completed with “sufficient” to pass the course and take the final exam.*



Assignment Teams

Each consists of 3-5 students

Teams are formed on CANVAS after confirming the actual number of attending students (second week)

Teams remain the same during all main and mini assignments



* The presentation of mini & main assignments will start on week 3. Thus you will have enough time to meet/mingle with your team members.

Course Material

- **Lecture Slides:** shared each week before the lecture (obligatory for exam)
- **Tutorial Slides:** shared each week before the lecture (obligatory for exam)
- **Journal Articles:** shared on blackboard (all are obligatory for the exam/ see course outline)
- **Guest Lectures:** content and main learnings (helpful through not obligatory for the exam)

FAQ Course Material:

- Other book/s and recommended readings (NOT required for the exam – helpful for further learning)
- ARTICLES You don't need to know everything about each article. Only MAIN LEARNINGS (objective, main results and conclusions) are essential for the exam and those could mostly be summarized in few sentences, few bullet points or 1-2 figure. Try to get the essence of each article.
- GOOD NEWS is that, these main learnings from articles are mostly and already summarized in lecture slides in each (related) lecture !

Digital Landscape

Digital Marketing & Analytics: Value & Motivation



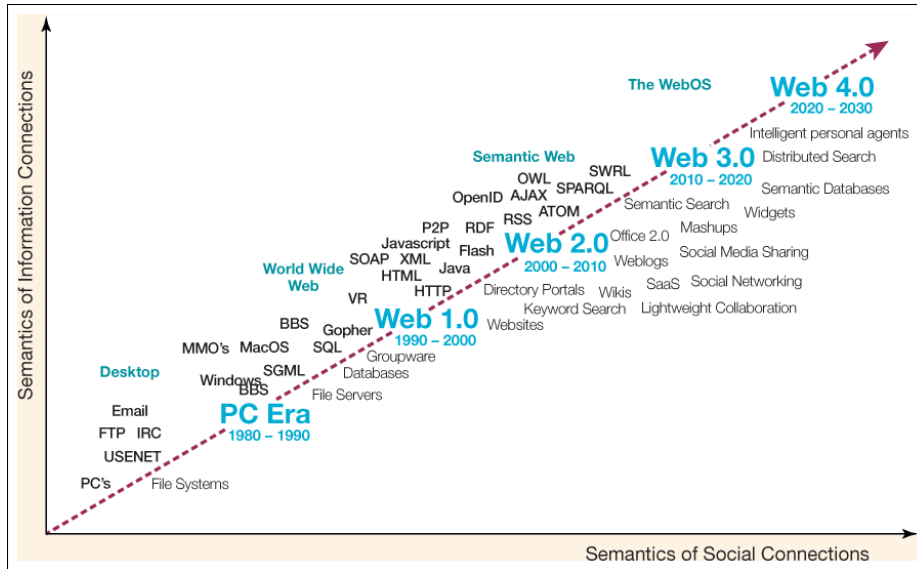
"Half the money I spend on advertising is wasted; the trouble is, I don't know which half." John Wanamaker, CEO, Wanamaker's Department Stores

Main Challenge: Measuring the real IMPACT – no speculation, no wishful-thinking – we need to know what really matters

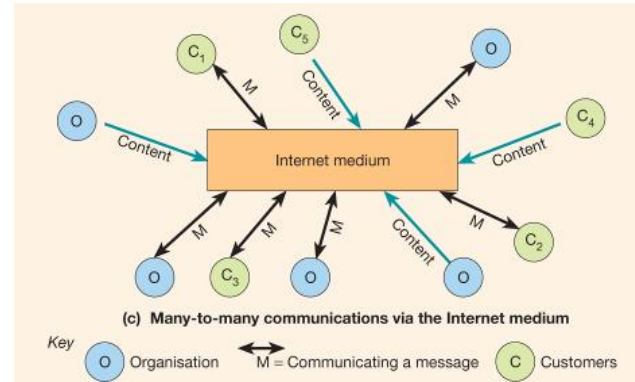
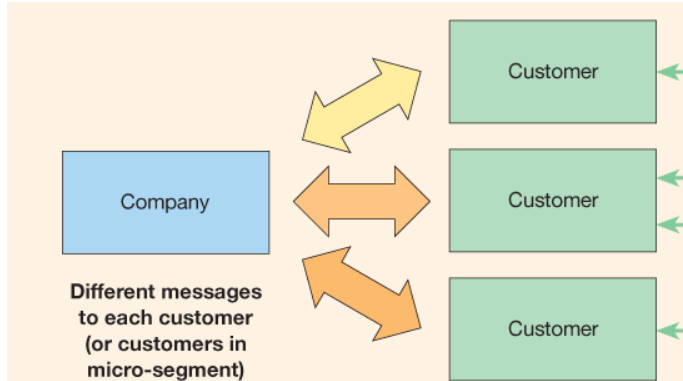
Digital Landscape: Complex & Expanding

Online data getting more and more complex and challenging now

But the data is also getting rich and diverse to have a better understanding and then, predict and manage customers' behavior.



Digital is Distinct: Personalization & User Generated Content



Personalization (Individualisation) and UCG what make today's digital platforms substantially different than traditional offline marketing

- TV, Radio, Newspaper: (non)digital traditional media (same message to all);
- New Digital media (unique-personalized messages and more information exchange between customers)
- Customers actively generate content and impact/shape-up other customers and firms' behavior

Digital Marketing: Customer Journeys & Touchpoints

All these goals and benefits can be achieved through designing and managing **seamless customer journeys** on digital platforms – by using **right digital touchpoints synchronized with offline platforms & touchpoints**.



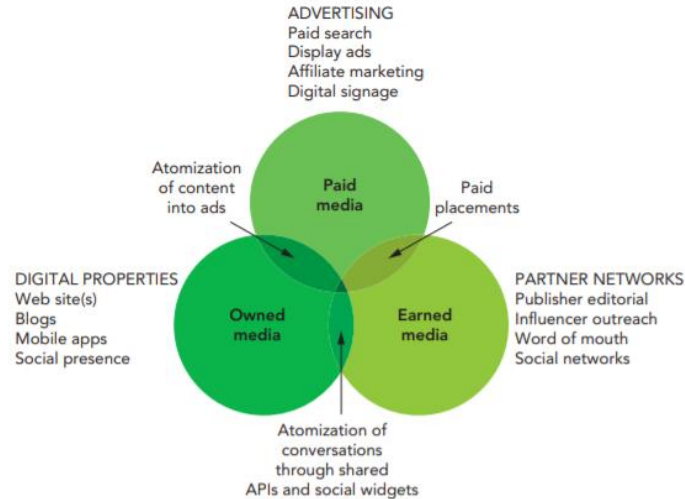
Then let's start by taking a look at: what are these TOUCHPOINTS

Touchpoints:
Paid-Owned-Earned Media

Paid-Owned-Earned Media

Paid-Owned-Earned Touchpoints: Most widely used classification in marketing & advertising practice.

Know and used by most marketers. NOTE: This is not a must-use & universal recipe, but easy to understand and implement.



	PAID	OWNED	EARNED
The essence?	Advertising	Content	Engagement
Who initiates a contact?	Brand	Customer (if brand is prepared)	Customer
Form of communication?	Brand to Customer	Interactive between brand and customer	Customer to customer

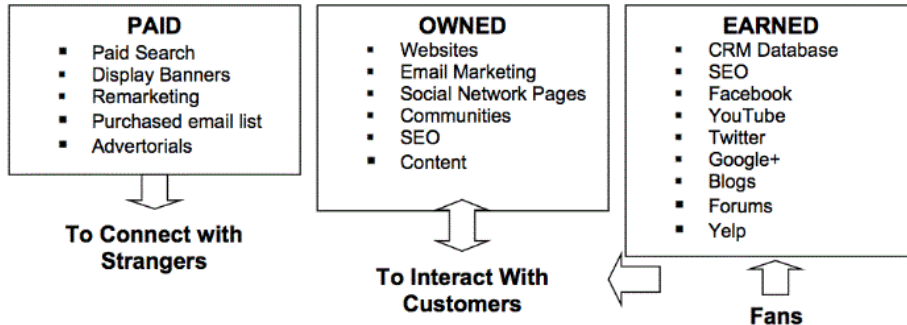
Paid-Owned-Earned Media: In a Nutshell

Owned Media, i.e. your web site, your app, your social media platforms – so where you can post your own information for ‘free’.

Earned Media is the most difficult, yet a valuable type of media. Word of mouth, conversations, comments, likes and shares.

Paid Media: where you pay including > advertising, search engine advertising and promoted social media content.

Thus a vlogger or influencer is by definition Paid Media (NOT Earned) since it practically is a paid medium. However – going one step beyond – the customer-to-customer sharing stimulated by influencer, then is an earned activity.



Paid-Owned-Earned Media: Relevance

Why do we need such classifications or why don't we handle each touchpoint in a granular way ?

Hundreds of online (and offline touchpoints) – hard to make managerial decisions for each touchpoint.

POE helps decision making whether the firm should focus on advertising (paid), content & service (owned) or engagement (earned)

Paid	Owned	Earned
Online Ads Banner Ads in Websites Banner Ads in Mobile Apps Online Pop-Up Ads Campaign Messages (email) Product / brand placement on online platforms Online Sponsorships Paid Search (SEO)	Firm (own) website Supportive (own) websites Mobile applications of the brand Social media accounts of the brand/company	Social WOM Online Reviews Articles, interviews about the brand in online media Blogs, forums, user evaluations Non-Paid Search Results

POEM: **Paid**
Owned

Earned

FOOD FOR THOUGHT:
A firm can also earn **NEGATIVE** engagement. Thus earned does not necessarily mean positive!
I.E: Negative word-of-mouth (rumors)

Paid-Owned-Earned Media: Extension > Category Media

Extension on Paid-Owned-Earned framework:

它包括两大类:

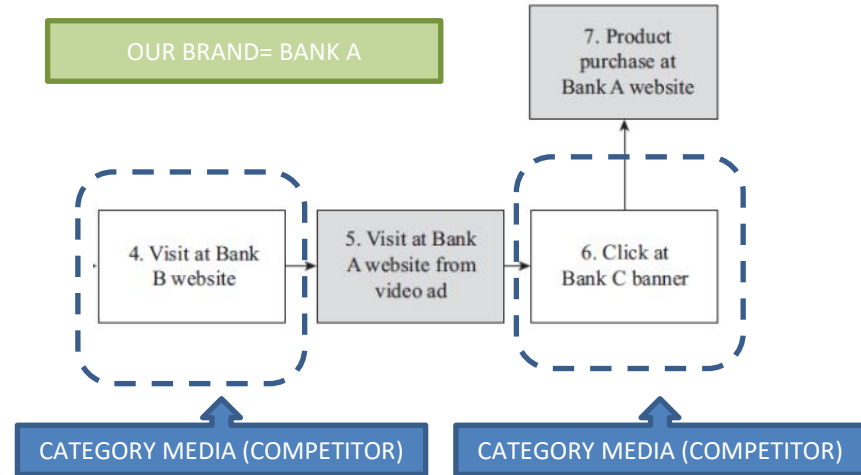
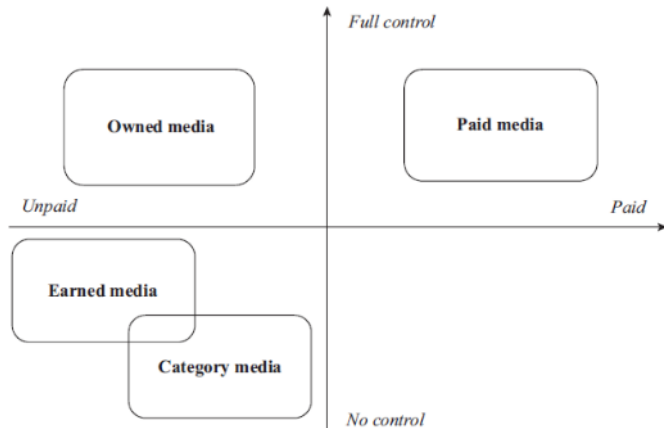
竞争对手的媒体内容 (包括他们的自有媒体、付费媒体、赢得媒体);
独立媒体的类别内容 (例如消费者评测网站、行业新闻、产品对比文章等)。

Category Media: relates to the category that do not mention the advertiser's brand neither published by the focal brand.

Category Media (类别媒体) 指的是那些没有提及你品牌、也不是由你品牌发布的内容, 而是与整个产品类别有关的内容。

Captures (i) competitors' own, paid, and earned media and (ii) independent publications related to the product category.

Motivation: Control Level on Earned (Individuals) and Category (publications, competitors) different. Control: Earned > Category



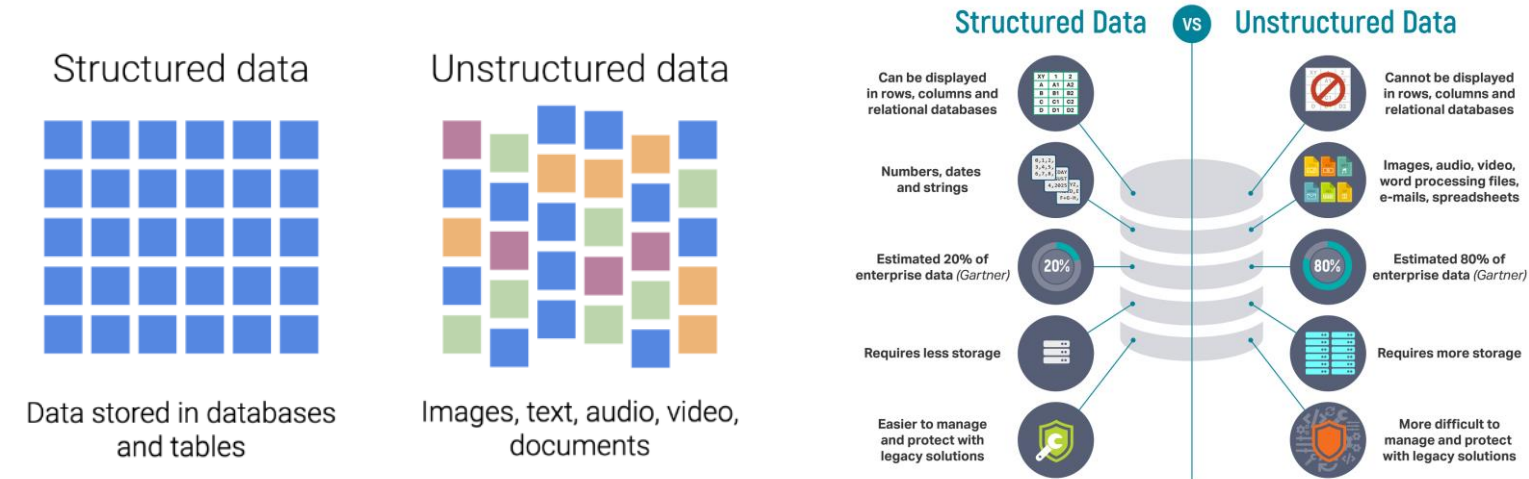
Digital Data Types & Metrics

Digital Data Types: Classification 1 – Structure

Data could be structured vs unstructured in nature. The management and analysis of the data depends on this structure accordingly.

Structured Data: Mostly Quantitative (numeric). Examples: Sales in Euro, Click/No Click (0-1), Session time, Number of Likes etc.

Unstructured Data: Mostly Qualitative: Textual or Visual Data: Photos/Videos, Examples: Instagram Videos, Online Reviews, Social Media Posts



Digital Data Types: Classification 1 – Structure

Here is a simple example for you to better understand how structured, unstructured or semi-structured the same data could be:

Unstructured data

The university has 5600 students.
John's ID is number 1, he is 18 years old and already holds a B.Sc. degree.
David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.

Semi-structured data

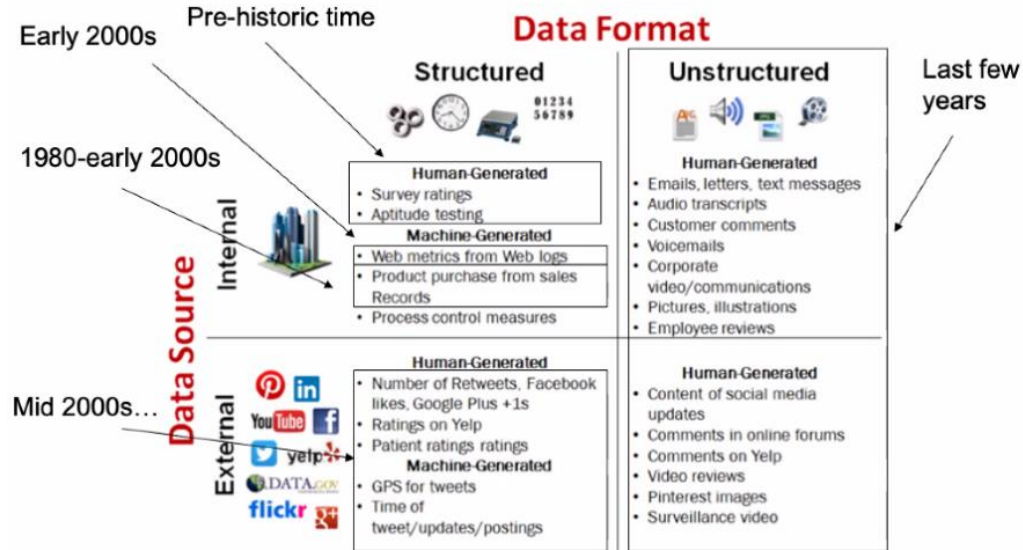
```
<University>
  <Student ID="1">
    <Name>John</Name>
    <Age>18</Age>
    <Degree>B.Sc.</Degree>
  </Student>
  <Student ID="2">
    <Name>David</Name>
    <Age>31</Age>
    <Degree>Ph.D. </Degree>
  </Student>
  ....
</University>
```

Structured data

ID	Name	Age	Degree
1	John	18	B.Sc.
2	David	31	Ph.D.
3	Robert	51	Ph.D.
4	Rick	26	M.Sc.
5	Michael	19	B.Sc.

Digital Data Types: Classification 1 – Structure

Unstructured Data is now more prominent. Projections: unstructured data is over 80% of all business-marketing data.



Digital Data Types: Classification 2 – Source of the Data

- (a) **On-Site Data:** what you see on website: available and visible to anyone. extracted through: data/web scraping
- (b) **Clickstream/Session Data** traffic on website-app: what, when and how people do online – not visible to visitor (i.e: Google Analytics)



Data Source: (1) On-Site Data - Web Scraping

This is what you see on the website (product lists, comments, reviews etc) – visible to all viewers & visitors.

Could be structured (i.e: number of likes) or unstructured (i.e: user reviews in tekst)

COLLECTED BY WEB SCRAPING: Collecting data from web sites, social media platforms (i.e: such as product information/listings, customer reviews, ratings, comments, posts through scraper-software and importing to local data base in xls, sql or other format.

The screenshot shows the Yelp website interface for searching restaurants near Washington, DC. The search bar at the top is set to 'Near Washington, DC'. Below the search bar, the results are filtered by 'Best Restaurants in Washington, DC'. A list of restaurants is displayed, with the first two being 'DuPont Circle' and 'The Bird'. Red boxes and arrows highlight specific data points: 'City/Locality' (Washington, DC), 'Business Name' (DuPont Circle), 'Address' (1361 Connecticut Ave NW), 'Price Range' (\$\$), 'Number of Reviews' (200), and 'Category' (Desserts, Cafes, Bakeries). A map on the right shows the location of the restaurants in Washington, DC.



business_name	business_url	business_phone_number	business_address	business_rating
Meier Coffee	https://www.yelp.com/f/ (647) 350-9120		1040 St Clair Avenue W	4.5 star rating
Fuwa Fuwa Japanese Pancakes	https://www.yelp.com/f/ (647) 618-2888		408 Bloor Street W	4 star rating
Fahrenheit Coffee	https://www.yelp.com/f/ (647) 896-1774		120 Lombard Avenue	4.5 star rating
Quantum Coffee	https://www.yelp.com/f/ (647) 494-4429		460 King Street W	4 star rating
Black Canary Espresso Bar	https://www.yelp.com/f/ (416) 301-7633		329 Yonge Street	4.5 star rating
The Coffee Lab	https://www.yelp.com/f/ (647) 212-8817		141 Spadina Avenue	4.5 star rating
Strange Love	https://www.yelp.com/biz/strange-love-toronto-2?osq=Coffee+Shop		101 Spadina Avenue	4.5 star rating
Voodoo Child	https://www.yelp.com/f/ (647) 748-1011		388 College Street	4 star rating
Himalayan Coffee House	https://www.yelp.com/f/ (416) 485-6464		2552 Yonge Street	4.5 star rating
Forget Me Not Cafe	https://www.yelp.com/f/ (416) 848-0057		506 Adelaide Street W	4 star rating
Dineon Coffee	https://www.yelp.com/f/ (416) 900-0949		140 Yonge St	4 star rating
Shy Coffee	https://www.yelp.com/biz/shy-coffee-toronto?osq=Coffee+Shop		3-766 King Street West	4.5 star rating
Piedmont Coffee Bar	https://www.yelp.com/f/ (416) 901-8639		66 Isabella Street	4.5 star rating
Jimmy's Coffee	https://www.yelp.com/f/ (416) 792-1141		84 Gerrard Street W	4 star rating
Snap Coffee and Tea	https://www.yelp.com/f/ (647) 350-8268		100-86 Gerrard Street E	4.5 star rating
Rooster Coffee House	https://www.yelp.com/f/ (416) 995-1530		343 King Street E	4.5 star rating
Boxcar Social	https://www.yelp.com/f/ (844) 726-9227		1208 Yonge Street	4 star rating
NEO COFFEE BAR	https://www.yelp.com/f/ (647) 348-8811		161 Frederick St	4 star rating
Rialto Espresso Bar	https://www.yelp.com/f/ (647) 344-6566		266 Adelaide Street W	4.5 star rating
Building Coffee	https://www.yelp.com/f/ (416) 606-2275		100-86 Gerrard Street E	4.5 star rating
HotBlack Coffee	https://www.yelp.com/f/ (416) 840-6141		245 Queen Street W	4.5 star rating
Carbonic Coffee	https://www.yelp.com/f/ (647) 352-8550		37 Baldwin Street	4 star rating
Sorry Coffee	https://www.yelp.com/biz/sorry-coffee-toronto?osq=Coffee+Shop		102 Bloor Street W	4 star rating
Moon Bean Coffee Company	https://www.yelp.com/f/ (416) 595-0327		30 St. Andrew Street	4 star rating
Fahrenheit Coffee	https://www.yelp.com/f/ (647) 896-1774		529 Richmond Street W	5 star rating

TUTORIAL 2 (10 NOVEMBER) to see and practice how we do that! Note: Required in mini data assignments !

Data Source: (2) Clickstream-Session Data

CLICKSTREAM = BEHAVIORAL DATA (session behavior)

This is NOT a data on what is visible on the website, but **what customer do/ what they click** on the website !

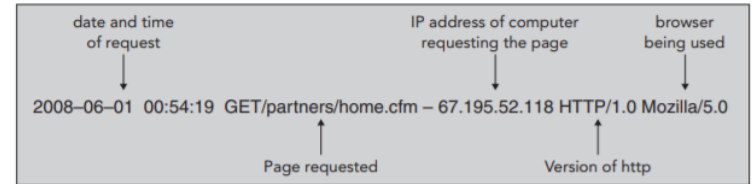
COOKIE : Tiny text file in your web browser storing information about your browsing experience (e.g. login information, user preferences, shopping cart) > SOURCES: Google Analytics (now GA-4), BigQuery etc.

1 Individual Level Data: recording each visitors' individual behavior with a time-stamp. i.e what each customer does step by step.

2 Aggregate Data: total numbers. not focusing on individual level behavior instead > total number of visitors etc.

The screenshot shows a text file named 'Omniture_0.tsv.gz' with columns for various user and session data. Red arrows point from labels to specific columns: 'Timestamp' points to the date and time column, 'Registered User SWID (if logged in)' points to the user ID column, 'IP Address' points to the IP address column, and 'Geocoded IP Address' points to the geocoded IP column. A red box highlights a row of data.

Registered User SWID (if logged in)	Timestamp	IP Address	Geocoded IP Address	URL
1331799426	2012-03-15 01:17:06	2060005755985467733	4611687631106657821	FA5-2,8-A53
N 0	99.122.210.248	10	http://www.acme.com/SH55126545/VD5517036	
4	{7AAB8415-E803-3C5D-7100-E362D7F67CA7}			
U	en-us,en;q=0.5	516	575	1366
N Y	2 0	304	sbglobal.net	15/2/2012 4:16:0 4 240 45 41 10002,00
011,10020,00007	Mozilla/5.0 (Windows; U; Windows NT 6.1; en-US; rv:1.9.2) Gecko/2010115 Firefox/3.6			
48 0 0	2 3 0	homestead	usa	528 fl
0		0		0 0 0



Data Source: (2) Clickstream-Session Data

GOOGLE ANALYTICS (CURRENTLY GA-4)

Most common tool in the market, Free (in return you hand over your data to Google)

Good enough for most small/medium companies.

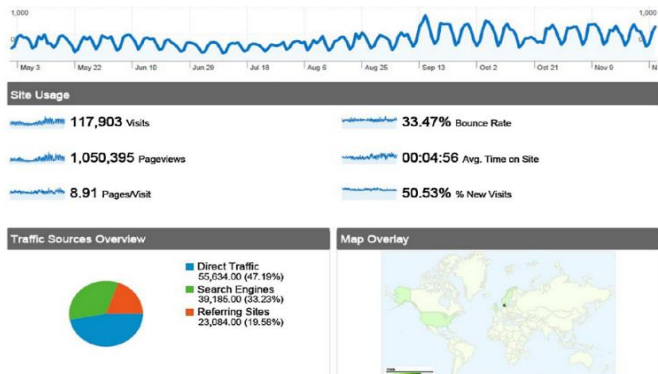
Aggregated data (limitation if you want to do advanced analysis) Other options available for more demanding data scientists.

In new generation GA-4 it is also possible to view individual-session level data.



Device category [?]	Sessions [?] ↓	% New sessions [?]	New users [?]	Bounce rate [?]	Pages/ session [?]	Avg. session duration [?]
	36,923 % of total: 100.00% (36,923)	71.45% Avg. for view: 71.36% (0.13%)	26,383 % of total 100.13% (26,350)	66.19% Avg. for view: 66.19% (0.00%)	6.55 Avg. for view: 6.55 (0.00%)	00:03:33 Avg. for view: 00:03:33 (0.00%)
1. Desktop	29,707 (80.46%)	72.85%	21,642 (82.03%)	65.18%	7.61	00:03:53
2. Mobile	5,565 (15.07%)	70.67%	3,933 (14.91%)	73.53%	1.86	00:01:33
3. Tablet	1,651 (4.47%)	48.94%	808 (3.06%)	59.54%	3.20	00:04:10

Data Source: (2) Clickstream-Session Data



AGGREGATE
GOOGLE ANALYTICS
BASIC (FORMER) VERSION

- 1 How are websites found by the customers ?
- 2 What is the content used by visitors ?
- 3 How often do users come back to website?
- 4 How many new users are there?
- 5 What do we know about the users?
- 6 What devices are used to visit the websites?

3-05-24 06:20:49.098186 UTC	week	morning	non-working hc	1020566708	T	https://nl.pgonline.org/nl1-8/2utm_source	mobile	Zoetermeer		
3-05-24 06:20:49.098186 UTC	week	morning	non-working hc	1020566708	B	https://nl.pgonline.org/nl1-boezamforleren	mobile	Zoetermeer		
3-05-24 06:20:49.098186 UTC	week	morning	non-working hc	1020566708	B	https://nl.pgonline.org/nl1-boezamforleren	mobile	Zoetermeer		
3-05-24 06:20:49.098186 UTC	week	morning	non-working hc	1020566708	B	https://nl.pgonline.org/nl1-boezamforleren	mobile	Zoetermeer		
3-05-23 09:18:37.734691 UTC	week	morning	working hours	1021178927	T	https://nl.pgonline.org/nl1-12/2bclid=wa	desktop	Fort Worth	(referral)	
3-05-23 20:17:17.873689 UTC	week	evening	non-working hc	1020087284	P	https://nl.pgonline.org/nl1-forum/onderwerp	mobile	The Hague	(organic)	
3-05-23 11:54:02.09827 UTC	week	morning	working hours	1027267338	H	https://nl.pgonline.org/nl1-artikelen/kan-een	mobile	Bovenkarspel	(direct)	
3-05-24 02:01:59.157989 UTC	week	night	non-working hc	102770642	V	https://nl.pgonline.org/nl1-forum/onderwerp	mobile	Hellevoetsluis		
3-05-24 05:29:40.287298 UTC	week	night	non-working hc	102810737	2	T	https://nl.pgonline.org/nl1-8/2utm_source	mobile	Leerdam	
3-05-24 00:10:25.990852 UTC	week	night	non-working hc	1028204440	T	https://nl.pgonline.org/nl1-8/2utm_source	mobile	Amsterdam		
3-05-24 08:16:24.655795 UTC	week	morning	non-working hc	1030099052	S	https://nl.pgonline.org/nl1-forum/onderwerp	desktop	Leek		
3-05-23 18:15:57.184452 UTC	week	evening	non-working hc	1036375916	B	https://nl.pgonline.org/nl1-forum/onderwerp	desktop	Billhoven	(organic)	
3-05-24 05:11:06.680626 UTC	week	night	non-working hc	1047539598	T	https://nl.pgonline.org/nl1-8/2utm_source	tablet	Rotterdam		
3-05-23 17:12:53.200626 UTC	week	afternoon	working hours	1062048982	H	https://nl.pgonline.org/nl1-artikelen/kan-een	mobile	Haarlem	(direct)	
3-05-23 22:18:04.921238 UTC	week	evening	non-working hc	1068233772	V	https://nl.pgonline.org/nl1-artikelen/verband	mobile			
3-05-23 13:44:26.306287 UTC	week	afternoon	working hours	1069128085	T	https://nl.pgonline.org/nl1-8/2utm_source	mobile	Wagrum	Whello Phase	
3-05-24 10:34:13.934289 UTC	week	morning	working hours	1074692646	B	https://nl.pgonline.org/nl1-forum/onderwerp	desktop	Zoersel		

SESSION LEVEL
GA-4 & BIGQUERY

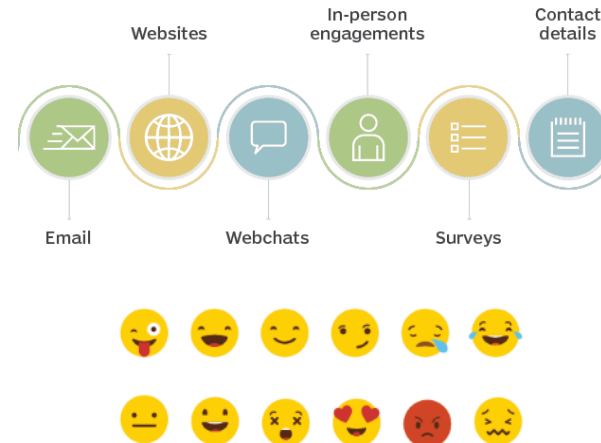
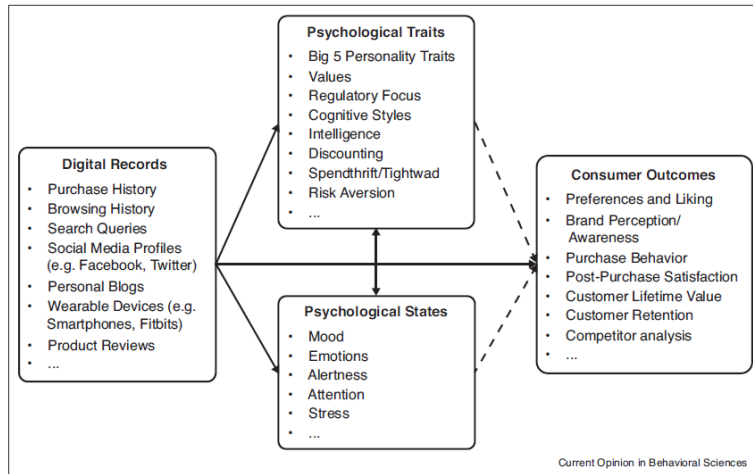
Digital Data with Emotions and Behavior

Digital data can also include (i) business outcomes and (ii) customer attitudes/emotions

Emotions and attitudes are non-observable. However this link can be built by:

1 Integrating survey (online marketing research) data with click-web data and business outcomes

2 Deep diving on possible emotional outcomes: Example: sentiment analyze of customer reviews, analyzing EMOJIS on social media



Finally: First, Second and Third Party Data

Below is the main mechanism of collecting/extracting 1. 2. and 3. party data from related platforms.

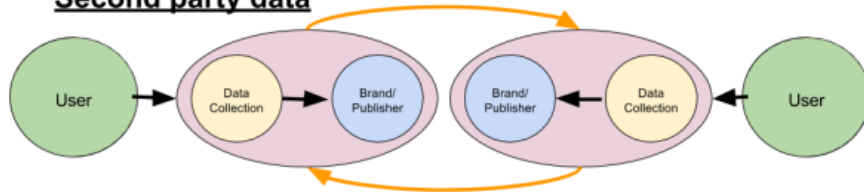
From a broader perspective (3.) Third Party data does not have to be sold/ commercialized but it also includes other 3. party public data-sets.

First party data



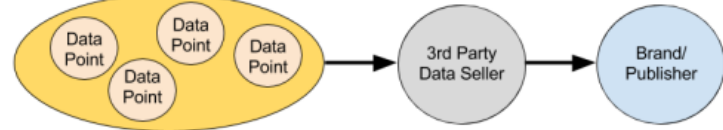
First party data is owned by brand/publisher. The data is collected by the brand/publisher directly from their users.

Second party data



Second party data is first party data that can be purchased or traded with another brand/publisher. Marketers/Publishers can go direct to companies they know will have the most relevant data, which is missing in their own first party data.

Third party data



Third party data is collected by data collection vendors from disparate data sources and sold to publishers/brands to use in campaigns and augment existing data.

Data: First, Second, Third (and Zero) Party Data

FIRST-PARTY DATA

Firm obtains data directly from/with its own sources, the company owns it. Significantly valuable because you get your information firsthand from your consumers, eliminating any misinterpretations and errors. It is, by far, the most effective and reliable form of data collection.

Some examples of First Party Data:

- Google Analytics
- Customer surveys, feedback, interviews, etc.
- Website

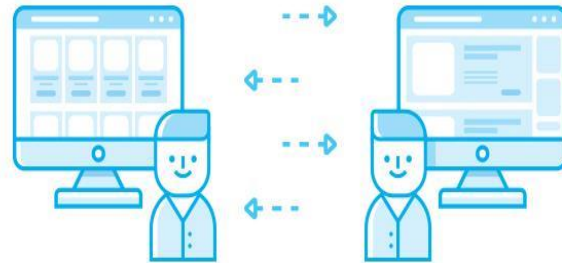


SECOND-PARTY DATA

Between first and third party data: Customized third-party data. Data is obtained from reputable, trustworthy third-party data sources and customize it for a specific target (from Google, or Facebook)

Some examples of Second Party Data:

- Google Big-Query (360) Attribution Data (through partnership)
- Facebook campaign specific data (through partnership)
- Data media publishers sell to advertisers (through partnership)



Data: First, Second, Third (and Zero) Party Data

THIRD-PARTY DATA

Any data firm obtains that is not its own (or customized).
The collected information is generally the same, but the third party does the work.

Here are some examples of Third Party Data:

- Oracle Data Cloud
- Salesforce Marketing Cloud
- Google
- Facebook (general/not campaign)
- Amazon
- Visa



ZERO-PARTY DATA

Can be confusing because same as first-party data in many ways.
Difference: “data that a customer **intentionally and proactively shares with a brand**, include preferences, purchase intentions, personal context, and how the individual wants the brand to recognize her.”

ZERO PARTY	FIRST PARTY
Collected from the user	Collected from the user as part of a transaction or behavior analysis
Up to date, high quality data	Up to date, good quality data
Collected with consent	Collected with consent
Examples of data: Communication preferences Interests & hobbies Purchase intents	Data Examples: Email Billing information Shipping address Most visited pages

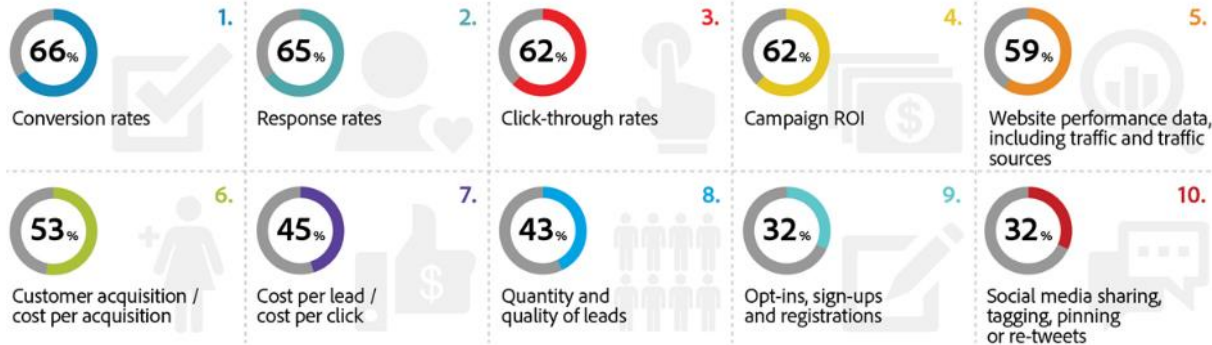
Metrics/KPIs on Digital Platform

Digital Marketing Metrics

Marketing Metrics: Tools helping companies **quantify, compare, and interpret** their own performance from marketing activities (Kotler & Keller, 2007).
In other words: Measuring the **impact** of marketing activities

The most important ones are referred to as **key performance indicators (KPIs)**.

Top 10 Metrics used by marketers:



Digital Marketing Metrics

What are the best metrics?

The answer is – There is nothing such as (per-se) best metrics. The decision on which one(s) to use depends on your objectives.

- Do you want to monitor/increase the engagement?
- Do you focus on increasing your sales (conversions)?
- Do you wish to enhance the awareness of your brand?



Search Engines

- % Click-through rates
- Backlinks
- Organic position for commercial keywords



Social Media

- Brand mentions
- Content shares
- Engagement



Email

- CTA click-throughs
- Forward rate
- Lead-to-customer conversion rate

Marketing Management (YOU) make the decision > Data Analysts execute the analysis.

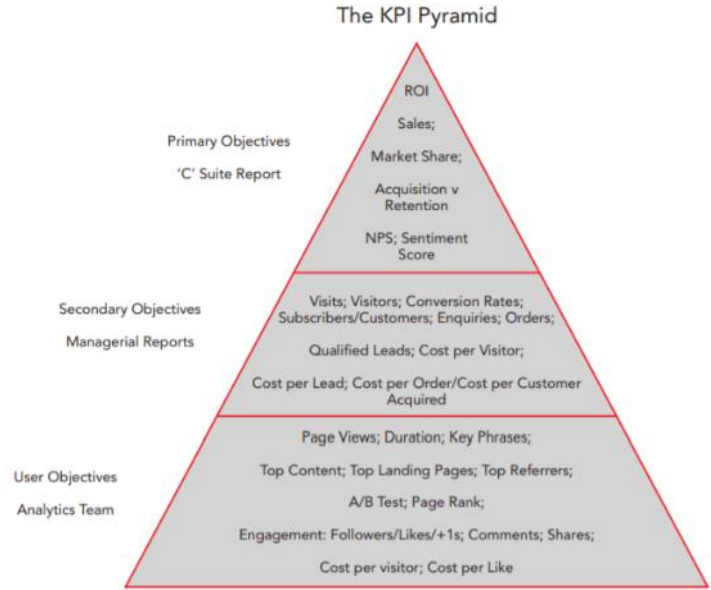


Figure 10.4 KPI Pyramid

Source: Originally by Pulizzi (2013) and adapted by Smith (2016)

Digital Marketing Metrics in Funnel: An Example

Which metrics to use > Correct question = Which stage of the funnel I am currently focusing on ?



Covering Digital Landscape: RACE Model



RACE marketing planning model is to provide a simple structure for companies to develop a omnichannel marketing communications plan which meets the challenges of reaching and engaging online audiences to prompt conversion to online or offline sales.

- **Reach:** Getting as many people as possible to know about your brand.
- Advertising online, posting on social media, ads in newspapers to attract people to visit your websites or social media pages.
- **Act:** Getting people who have noticed your brand to interact with it.
- They visit your website, check out your products, read your blog, or follow you on social media. You want them to take some action that shows they're interested.
- **Convert:** You're focused on making the sale (or in a broader sense conversion).
- You want the people who've shown interest in your brand to actually buy something, whether that's online through your website or in a physical store.
- **Engage:** Keeping your customers interested and active over the long term.
- After someone buys something, you keep in touch with them through emails, social media, or direct contact. You want them to keep coming back and to recommend you to others.

Digital Marketing Metrics: Stages of Customer Journey

Customer Journey has different stages from awareness-consideration (pre-purchase) to purchase and post-purchase (more in Lecture 2)

Each journey stage has certain metrics that are specific to that phase and more relevant in analytical-objectives.

Decision Stage	Need Recognition	Information Search and Evaluation	Purchase	Post-Purchase Evaluation
Types of Numeric Information	• Budgets	• Product Ratings • Product Rankings • Product Attribute Information • Price Information • Brand Identifying Information • Calorie Information	• Price Information • Promotion Information • Quantity Information • Time Information • Financial Information	• Consumer Ratings • Price Recall • Promotion Recall



Basic Metrics: Impression Rate & In-View Rate

Impression: % content is **displayed** once on a web page (the customer is **exposed to an online content** on a website)
Does **NOT** necessarily mean that the customer has actually seen the content though !

In-View Rate (%): % customer who actually **SEE** the online content he is exposed to.

Mostly below 50% of the impressions - people do not scroll down the page, or slips their attention (out of sight) or the viewer is not a actually human user

Media Rating Council (MRC) 50% or above actual (in-view) is sufficient; though this is still a debate ! (Fulgoni, 2016)

$$\text{Viewability Rate} = \left(\frac{\text{Total Measured Viewable Ad Impressions}}{\text{Total Measured Ad Impressions}} \right) \times 100$$

**Viewability rate is expressed as a percentage, so for easy of use x 100 is added to the above equation.*

What does it mean?

Total Measured Ad Impressions: Number of times an ad was loaded on a page (and counted by a server).

Total Measured Viewable Ad Impressions: Number of times an ad was loaded on a page (and counted by a server) which was 50 percent on screen for one second or longer (for video ads - two seconds or longer).

All three ads on this screen will count as viewable if they stay on screen for more than 1 second.



Only the top ad counts as viewable (if seen for at least one second) unless the user scrolls down.

Basic Metrics: Click Through Rate (CTR)

Most commonly used metric in practice by marketers:

Early days of online advertising (nineties): 1-2% - sometimes went up to 3% for successful campaigns. TODAY: went down to mostly below 0,1-0,2 %. Thus: **Less than 1 in 1000 impression** can generate a click !

CTR: Simple, fast and easy to measure. BUT not a good indicator marketing effectiveness (Fulgoni, 2016) Mostly used if we want to measure awareness, engagement or we are not able to measure conversions.

Click-Through Rate Formula

Calculate how frequently people click on your ads.

$$\text{CTR} = \left(\frac{\text{Total Measured Clicks}}{\text{Total Measured Ad Impressions}} \right) \times 100$$

(Click-Through Rate)

*Click-Through Rate is expressed as a percentage, so for easy of use x 100 is added to the above equation.

Multiplied by 100 for easiness of use since CTR % (actual) percentages are very low.

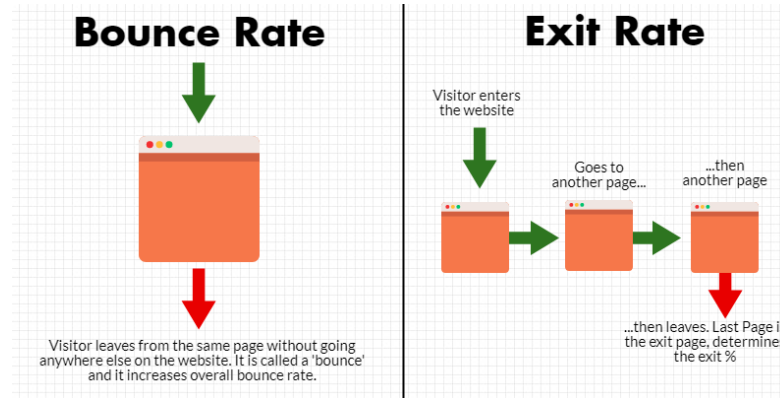
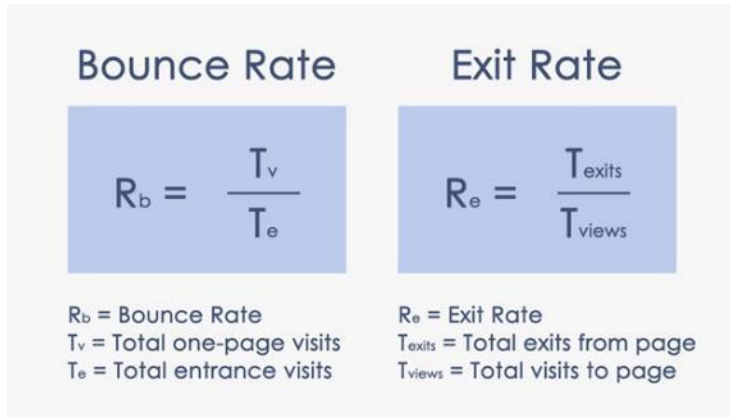
Basic Metrics: Bounce & Exit Rate

Bounce Rate: The (%) percentage of visitors to a particular website **who navigate away from the site after viewing only one page**.

Important to understand web page effectiveness/success and engagement

Bounce = Not per-se a bad thing: May mean your page is **not engaging** – But can also mean your customers are **able to find quickly what they are looking for** while searching for a product or information.

Some thinking: How would you interpret a HIGH Bounce rate but LOW Exit rate on your website ?



Conversion-Attribution

CTR and Conversion Rate: Fundamental Metrics

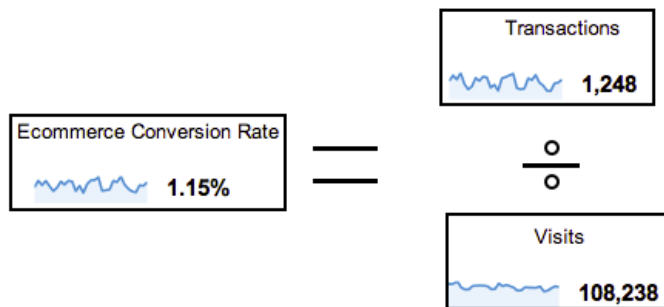
CLICK THROUGH RATE: number of clicks that your ad receives divided by the number of times your ad is shown: $\text{Clicks} \div \text{Impressions} = \text{CTR}$.
For example, if you had 5 clicks and 1000 impressions, your CTR would be 0.5%.

CONVERSION RATE: number of conversions divided by the number of total clicks (tracked) to a conversion during the same time period.
For example, if you had 50 conversions from 1,000 clicks: your conversion rate would be 5%, since $50 \div 1,000 = 5\%$.

Practical Tip – What Exactly is Conversion?:

Depends on your campaign objectives. In practice it is mostly the act of “purchasing” after certain engagement and clicks

But it could also be a subscription (an email group), registration (your website or app) or people watching your video (if that is your ultimate aim) !

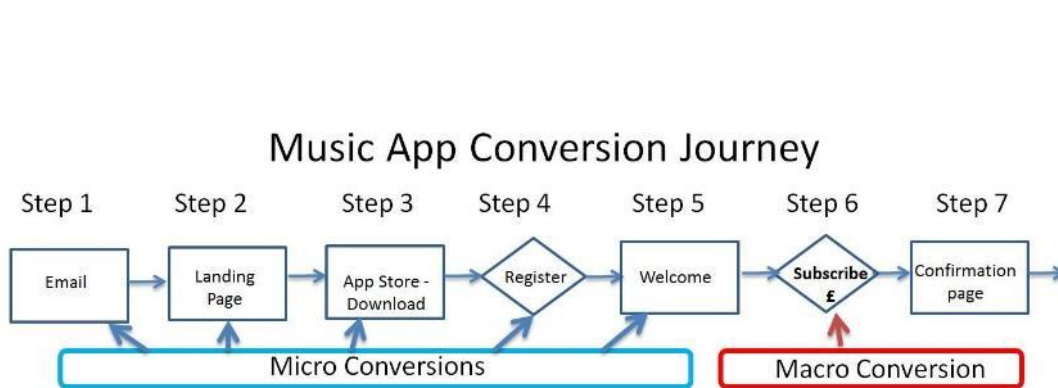


Conversion: Macro & Micro Conversions

Google names smaller engagements before purchases as “Micro-Conversions” – i.e. *Registering as a user before buying, watching product video*

Micro-conversions can have a lot more predictive power than one might expect.

For example, after building our model, we may realize **viewing all images** on a product page has big **predictive** power for **future purchases**.



Macro Conversion

A **macro conversion** is the primary **conversion** on a website, for example a completed sale on an ecommerce site or a completed lead generation form.



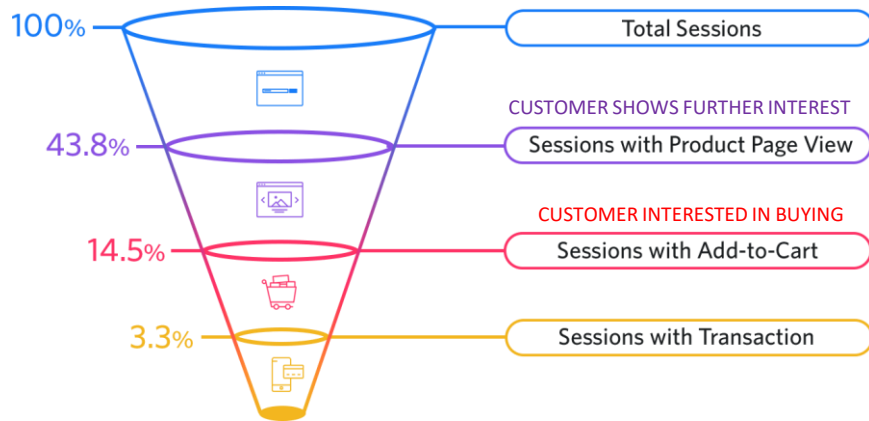
Micro Conversion

A **micro conversion** relates to smaller engagements such as a newsletter sign up or a user watching a product video. **Micro conversions** can often precede the **macro conversion**.

Conversion Funnel: A Fundamental Concept

A Conversion Funnel refers to different and multiple stages in a buyer's journey leading up to a purchase.

Depending on your campaign objective: marketing efforts can be aimed at "upper funnel", "middle funnel", or "lower funnel".



Which “Clicks” or Touchpoints are More Influential...on Conversion?

Assume we have 3 ads: 2% of the people that click on our CNN ad, 1.5% on our Facebook ad and 2.2% on our Google ad convert

What about people that saw all 3? We're counting the same conversion multiple times

In other words: For those who saw/click all these three ads – which ad/ads matter more – have the real influence on CONVERSION

How do we ATTRIBUTE the conversion over the 3 different channels? = **ATtribution PROBLEM**



Source: <https://www.thinkwithgoogle.com/tools/customer-journey-to-online-purchase.html#/the-us/travel/medium/generic-paid-search>

What would happen if we
Take this channel out of journey ?
Do we really need PAID SEARCH?

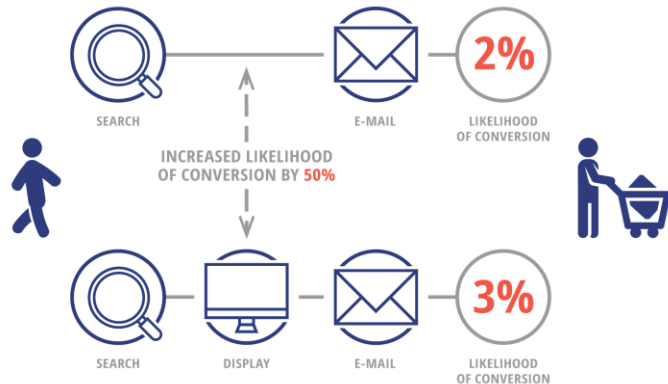
Does it really help to convert?
OR
Is it only costing us extra money?

Attribution: Finding Most Influential/Valuable Touchpoints on Customer Journeys

WHICH TOUCHPOINT IS MORE INFLUENTIAL-VALUABLE TO CONVERT CUSTOMERS?

The purpose is to quantify the influence each advertising impression has on a consumer's decision to purchase or convert.

In other words: WHICH media channels or advertising formats have **CONTRIBUTED TO A CONVERSION**.



What if there was no Display Ad in this journey?
What is Display Ads' net % contribution to conversions ?
Does it really worth to invest on that touchpoint?
The Answer: Attribution Models



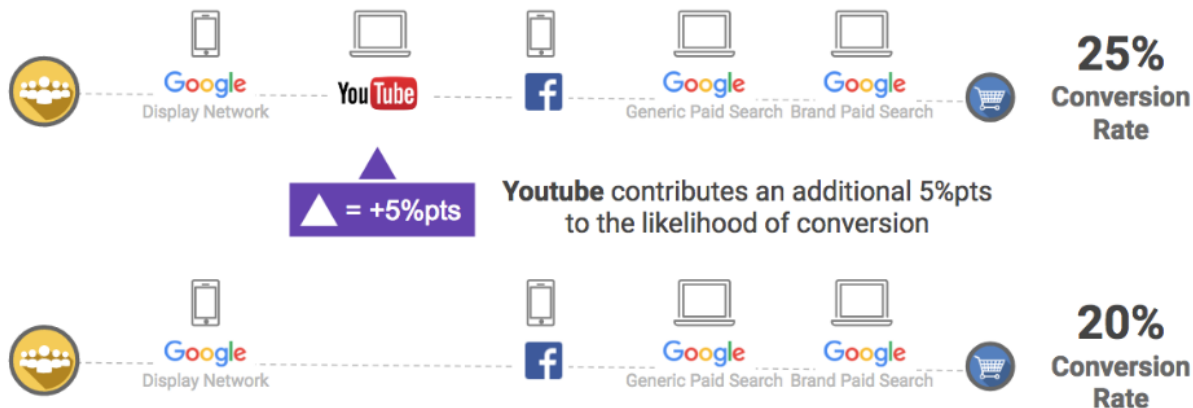
Given everything else is same Display Ads increase Conversion by 0.7%

Attribution: Conversion Impact of Each Touchpoint on Customer Journeys

Example: The differential (% conversion) impact of adding a Youtube Ad in customers' path to purchase.

Youtube Ad leads to 5% more conversion when included in customer journeys.

- What is the revenue impact of 5% more sales?
- What is the cost of adding a Youtube Ad to customer journeys?
- SIMPLE MATH = IF: Revenue increase with Youtube Ad (€) > The cost of adding the Youtube Ad (€) → USE Youtube Ad



Attribution: Conversion Impact of Each Touchpoint on Customer Journeys

Attribution is quantifying the VALUE of marketing ACTIVITIES with regard to DESIRED OUTCOME. **Three main types of attribution techniques:**

- 1 Simplistic (i.e last/first click)
- 2 Rule Based (Heuristic: Position Based, Time Decay)
- 2 Data-Driven (Algorithmic-Statistical)

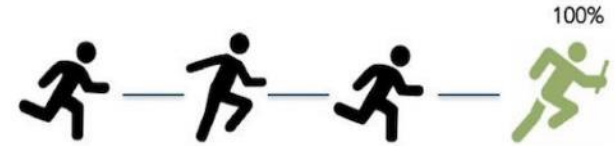


TYPE	NAME	DESCRIPTION
Single touchpoint	Last Interaction	Last click gets all the credit
	First Interaction	First click gets all the credit
Multi-touchpoint	Participation	All touchpoints get all the credit
	Linear (Even)	All touchpoints get equal credit
	Adjacency (Position)	Identifies the position away from the conversion; 1st (closest), 5th, etc.
	Latency	Identifies the lag time between the touchpoint and conversion
	Starter-Player-Closer (SPC)	Touchpoints get credit based upon custom weights for each position
	Custom	User-specified custom model (unique)
	Algorithmic	Statistics determine how much credit each touchpoint should get (unique)

Simplistic Techniques: Last-Click Inefficiency & Criticism

Industry Standard (Still): LAST CLICK WINS. Though receives many criticisms! (far from being logical)

Does your last-click necessarily have to be most influential on your purchase? Imagine your last shopping experience!



Think of media touch points in a consumer's path to conversion as a relay race team passing a baton to the finish line.

The last-click model would assign all credit for the race to the last relay racer who crosses the finish line, but would not give his teammates any credit for the win.

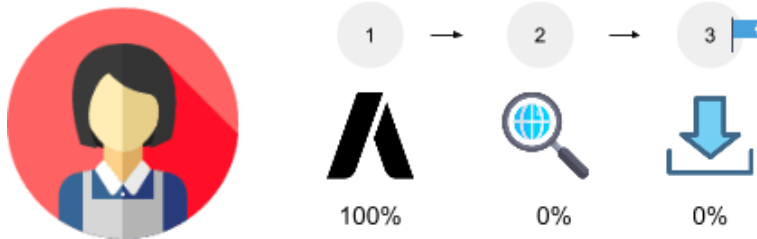
Simplistic Techniques: First Click Model

100% of the marketing success is attributed to the first known traffic source / referrer.
Thus all credits go to the JOURNEY STARTER touchpoint

Too much assumption based and biased;
However **could be useful** when the firm wants to **focus on the first touch points** with their brand, or if a firm especially focuses on building awareness on brands or product, highly value **starting new-journeys**.

WHICH touchpoints start the journey: Here focus is on new-more journeys; not on sales.

First Interaction Attribution Model



Rule Based Techniques: Uniformly Distributed (Linear) Model

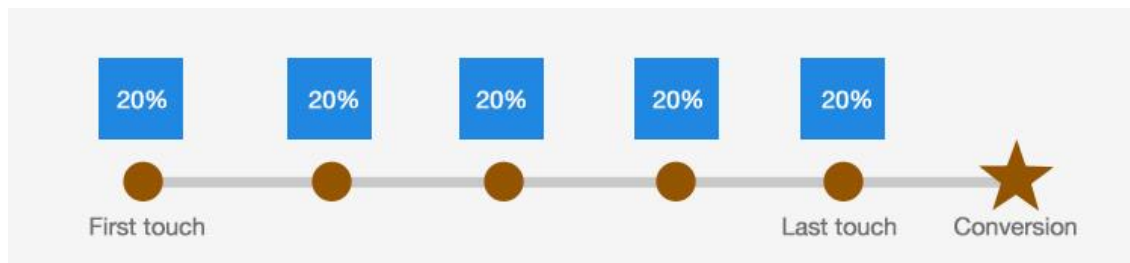
Claims all touchpoints along the journey **MATTERS – EQUALLY**.

The value of each conversion is distributed **uniformly (equally)** to all touchpoint encounters.

DRAWBACK: Because the shares of each conversion are divided equally among all channels. Model's assumption that each interaction in the customer journey has equal influence on the user's purchase decision thus is **NOT LOGICAL**.

When you plan to buy something do are clicks equally important/ remarkable for you? Probably NOT

USED: (Over)simplifies the attribution – when a fast/simple model needed taking all touchpoints in consideration



Rule Based Techniques: Time Decay Model

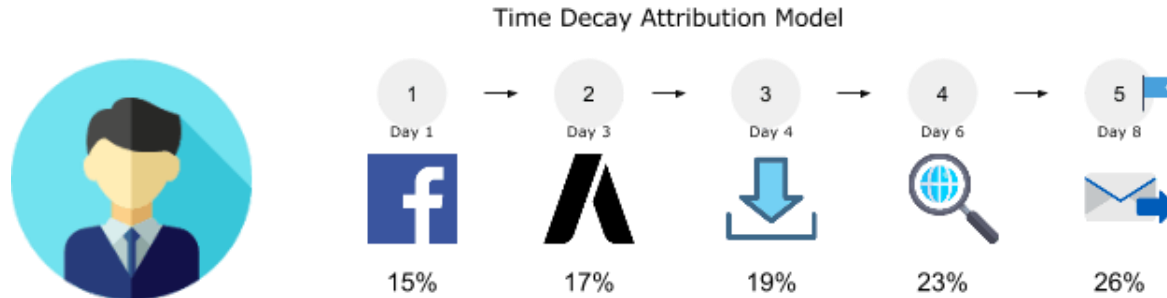
Time-decay model adjusts credit so that **the closer an impression is to a conversion, the more credit it receives**

ASSUMPTION on Memory Decay effect > More recent experiences = more influential

Drawback: touchpoints that close the deal benefit more from this attribution strategy, because the value is weighted progressively higher for channels nearest to the last impression. However a previous/ less recent experience moment could well be the main driver behind the conversion.

Aren't there decisive (maybe older) moments –not necessarily most recent - that are remarkable/convincing for you along your path to purchase ?

Mostly Used: for short-lived deals or promotional offers



Rule Based Techniques: Position Based Model

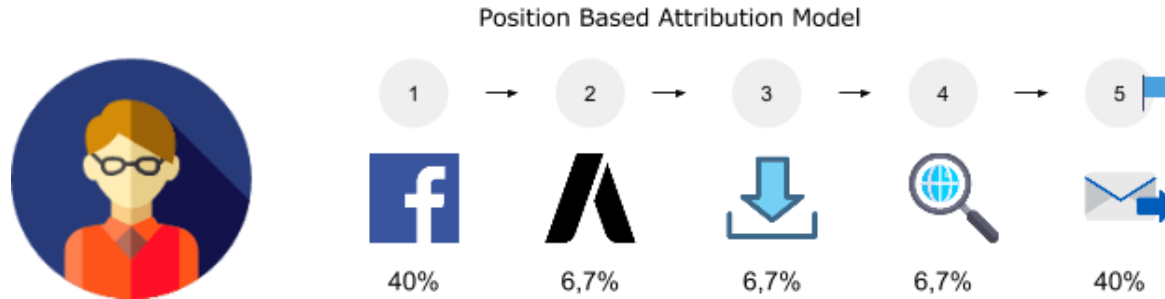
Uses a Pareto distribution rule, which attributes value to **specific parts of the customer journey**.

Places more importance on the **FIRST AND LAST TOUCH POINTS**. For example, under the 80/20 rule, 80 % of the conversion value is credited to the **FIRST AND LAST TOUCH POINTS**, and the remaining 20 percent is distributed to the other channels in the customer journey.

Assumes: First impression is important since it **attracts the user's attention**, and last one because of the **role it plays in concluding the transaction**.

What IF a moment in between (somewhere in the middle) might have influenced you the most to make your purchase decision ?

DRAWBACK: **biased** heavily toward the first- and last-click channels, arbitrary – assumption-based distribution of credits.



Source: Theory > Halgamuge & Kayande (2015) Nisar and Yeung (2017) Visuals: PiwikPro company website (2019)

Rule Based Techniques: Calculating Attribution Value

In rule (assumption) based simple techniques calculating attribution value is quite straightforward.

Let's say we have 7 customer journeys; among them 5 ends with CONVERSION

CLICKS ALONG THE CUSTOMER JOURNEY					Conversion Status
Start of the Journey				Most Recent	
C	A	D	E	A	YES
C	E	E	D	B	YES
B	E	C	D	B	YES
D	B	C	B	D	NO
E	C	D	E	B	YES
C	C	E	D	D	YES
A	B	B	C	E	NO

BE CAREFUL:

IN SIMPLISTIC/RULE BASED TECHNIQUES
YOU SIMPLY COUNT THE SHARE OF EACH TOUCHPOINT
IN CONVERSION (CONVERTED JOURNEYS)
DIFFERENCE WITH PROBABILISTIC TECHNIQUES:
THIS IS DIFFERENT THAN PROBABILISTIC TECHNIQUES
WHERE YOU CALCULATE THE MARGINAL CONTRIBUTION
OF EACH TOUCHPOINT IN A SEQUENTIAL PATH

STEP 1: You only take those journeys ending with CONVERSION = 5 journeys

STEP 2: You count (take a look at the share of each touchpoint) according to your technique.

Example: Last Click Technique > B is the most influential touchpoint (3/5) = %60 Attribution Value

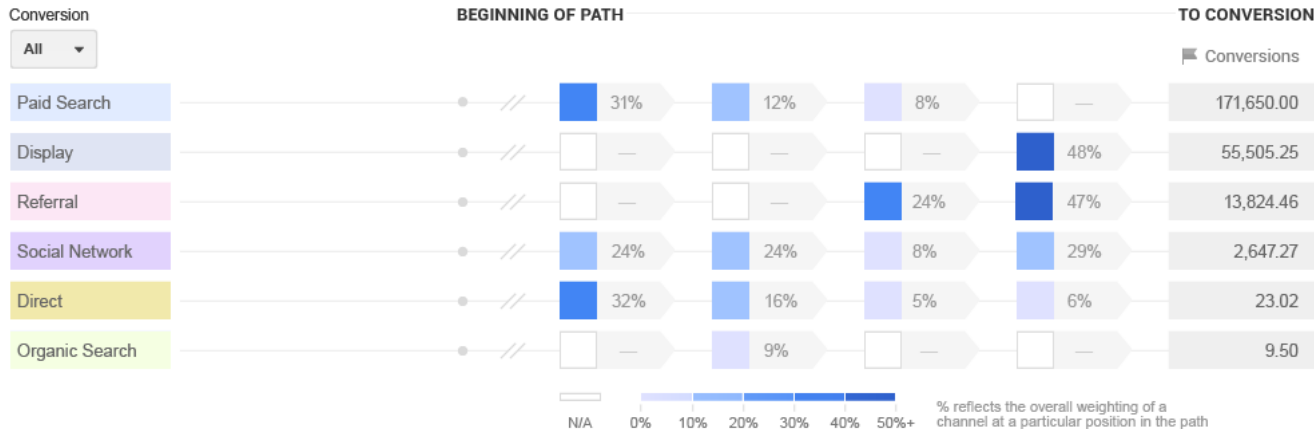
How about First Click Technique ?

Model Based & Data Driven Attribution Models

Now let's say: we want to know the **REAL VALUE** of each touchpoint without WISHFUL THINKING. In a way NOT BIASED by our ASSUMPTIONS

The better we predict conversion probability, the more we trust our parameters: We let the data tell us what is influential (no assumption)

Commonly Used: **Logistic Regression, Probabilistic Models, Game Theoretic Models (Shapley Value)**



Model Based & Data Driven Attribution: 1 Logistic Regression

Each observation is a click path that ends with a conversion or not

$Y = 1$ if conversion, 0 if no conversion

X = set of variables representing the path properties

different online touch-points; whether the customer encountered each of them or not (0 or 1), if so how many times (#) and when

$$P = \frac{e^{a+bX}}{1 + e^{a+bX}} \quad \text{Ln}\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

The β 's say something about the relative contribution of each touchpoint

For a practical example and how to interpret each click/touchpoint's effect see my data example during next **Tutorial on Thursday (29 October)!**

Note: Logistic Regression model equations goes beyond the scope of this course. In other courses such as Quantitative Marketing this is discussed in detail.



Model Based & Data Driven Attribution: 1 Logistic Regression

- # visits
- # page views per visit
- time since last visit
- per channel/ touchpoint:
 - # impressions
 - # clicks

PROS

Easy to use and interpret
 Insights on channel effects
 Additional explanatory variables can easily be added (i.e. time on site)

CONS:

Does not account for touch-point order, example:
 Effect of total # AdWords clicks $\vee$
 Effect of AdWords at begin or end of path X

Display	Search	Mobile	Email	Social	Convert?
1	1	0	0	1	1
1	0	1	1	1	0
0	1	0	1	0	1
0	0	1	1	1	0

SIMPLE LOGISTIC-REGRESSION ATTRIBUTION MODEL

$$X1 + X2 + X3 + + Xn$$

Touchpoint 1
Display

Clicked=1 YES
Not clicked=0 NO

Touchpoint 2
Mobile

Clicked=1 YES
Not clicked=0 NO

Touchpoint 3
Email

Clicked=1 YES
Not clicked=0 NO

Touchpoint 3
Facebook

Clicked=1 YES
Not clicked=0 NO

=

Y

CONVERSION
PURCHASE

PURCHASED=1 YES
NOT PURCHASED=0 NO

CUSTOMER 1 

0

1

1

0

1 PURCHASE

CUSTOMER 2 

1

0

1

0

0 NO PURCHASE

CUSTOMER 3 

1

1

0

1

1 PURCHASE

Model Based & Data Driven Attribution: 1 Logistic Regression

Interpreting Effect Sizes in Logistic Regression based Attribution Model

Tutorial 1: we shall learn how to interpret them !)

Check the relatively high/strong effects (Exp(B) and B coefficients) of Organic Search, Price Comparison, Paid Search and Retargeting on Conversions

TABLE 6 Logistic Regression Results

Variable	B	EXP(B)	Marginal effects
Intercept	-4.245***	.005	
Organic search	.436***	1.762	.0138***
Display	.174***	1.241	.0014***
Paid search	.358***	1.489	.0127***
Price comparison	.472***	1.683	.0145***
Retargeting	.394***	1.365	.0022***
E-mail	.147*	1.126	.0013*
Social media	.135**	1.113	.0019**
Observations	783,564		

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

Model Based & Data Driven Attribution: 2 Probabilistic Models

What IF we add/drop certain touchpoints to a journey? What is the impact of adding NU.NL as an additional advertising outlet ?

There are unlimited number of “**possible paths**” Good news: Although there are unlimited number of journey scenarios – most customers experience/ have certain types of journeys

Used by Google: Accurate if we have large number of (sample) journeys – capturing different journeys - for measurement.



- Marginal contribution of nu.nl advertisement to path B:
- $CR_B - CR_A = 0.12\% - 0.1\% = 0.02\%$

Model Based & Data Driven Attribution: 2 Probabilistic Models

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
B	2000	60	0.0300
C	3000	120	0.0400
AB	350	11	0.0314
AC	300	12	0.0400
BA	250	8	0.0320
BC	200	14	0.0700
CA	150	7	0.0467
CB	100	5	0.0500
ABC	10	1	0.1000
ACB	20	3	0.1500
BAC	30	4	0.1333
BCA	40	3	0.0750
CAB	50	7	0.1400
CBA	60	5	0.0833

PROBABILISTIC ATTRIBUTION: AN EXAMPLE

Let's say we have different journeys where different touchpoints are used (A,B,C)

How do we calculate the contribution (attribution value) of each touchpoint in the journey:

FEW RULES TO KNOW :

- 1 Position (Sequence) of touchpoint matters in Probabilistic Attribution
- 2 Thus: we don't only look whether A is used or not used (seen) by customer in journey
- 3 We also look at its position and use it in our calculation:
I.E Whether A is used in the beginning #1, middle #2 or in the end #3 as the last touchpoint
- 4 This gives us a more precise/accurate picture on the real effect of each touchpoint
- 5 Then we know the value of each touchpoint depending on its position in a particular journey

Let's say we have an CAB journey: where customer first see C, then A and finally B

What is the attribution value of each touchpoint in this path (journey) ?

Practical Solution: TUTORIAL 1 (Thursday 04 April) !

Self-Study Exercise

Sign-Up for a Google Analytics account now. (It is FREE) Sign-In if you already have one.

If you don't have your own website: Use Google Merchandise Store Demo Data (Dashboard) – open access to everyone:

<https://support.google.com/analytics/answer/6367342?hl=en>

Check what kind of metrics they have to measure/ analyze the online activity

Check their dashboard/ reporting tools and the “**conversion (attribution) funnels**”

Put some thought on: which of them could be used to analyze the issues that we saw in our slides such as: POE media (this week's slides)

Are there any metrics from our lecture that still cannot be analyzed by using the built-in metrics at Google Analytics standard package?



Thank You